

## Project ATHENA – Day 9

### Theme: Who Are You Actually Trading Against?

Welcome back.

Last time, we built the mental model that **markets are continuous auctions** driven by information, expectations, supply and demand, liquidity, and human behavior.

Today we're answering an even more important question:

#### **When you buy a stock, who is on the other side of your trade?**

Because understanding your competition is the first step toward building a strategy that can consistently outperform.

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### Learning Objectives

By the end of today's lesson, you should be able to:

- Understand the major market participants.
  - Know what advantages each participant has.
  - Recognize why retail investors can still succeed despite institutional competition.
  - Define where ATHENA should compete.
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### Lesson 1: Every Trade Has Two Sides

Imagine you buy **100 shares of Infosys**.

Someone sold them to you.

One of you is making the better decision.

Your job isn't simply to predict whether Infosys will go up—it's to understand **why the seller was willing to part with those shares**.

That seller could be:

- A pension fund rebalancing its portfolio.
- A hedge fund taking profits.
- A retail investor panicking.
- An algorithm reacting to new data.
- An employee selling vested stock.

The price reflects all of these competing motivations.

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## **Lesson 2: The Main Market Participants**

Think of the market as an ecosystem.

### **1. Retail Investors**

Examples:

- Individual investors using brokers.
- SIP investors.
- Swing traders.

Strengths:

- Flexible.
- Can trade small-cap stocks.
- No mandate to remain invested.

Weaknesses:

- Emotional decisions.
- Limited access to information.
- Often chase recent performance.

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### **2. Mutual Funds**

Objective:

Manage long-term investments on behalf of investors.

Strengths:

- Large research teams.
- Access to company management.
- Deep fundamental analysis.

Weaknesses:

- Massive portfolios.
  - Hard to buy or sell quickly.
  - Benchmark constraints.
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### 3. Hedge Funds

Objective:

Generate absolute returns.

Strengths:

- Quantitative models.
- Alternative data.
- Fast execution.
- Sophisticated risk management.

Weaknesses:

- High competition.
  - Capacity constraints.
  - Complex strategies can become crowded.
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### 4. High-Frequency Trading (HFT) Firms

Objective:

Exploit tiny inefficiencies.

Examples of advantages:

- Microsecond execution.
- Co-located servers.
- Advanced market-making systems.

These firms are competing over milliseconds—not weeks.

**ATHENA will not try to compete here.**

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### 5. Market Makers

Role:

Provide continuous buy and sell quotes.

They help maintain liquidity and earn the bid-ask spread while managing inventory risk.

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### 6. Corporate Insiders

Executives and employees may buy or sell shares for many reasons.

Insider trading based on material non-public information is illegal in most jurisdictions, but disclosed insider transactions can still provide useful signals.

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### Lesson 3: Why Retail Investors Can Still Win

This is one of the biggest misconceptions.

People often believe:

"Institutions always win."

Not necessarily.

Institutions face constraints that individuals often don't.

For example:

- A ₹50,000 crore mutual fund cannot meaningfully invest in tiny companies.
- A fund manager may be required to stay mostly invested even during weak markets.
- Many institutions track benchmarks and cannot deviate significantly.

An individual investor with a well-designed system can:

- Wait patiently.
- Ignore market noise.
- Focus on niche opportunities.
- Hold cash when risk is high.

These are real advantages.

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### Lesson 4: Where ATHENA Will Compete

This is a strategic decision.

ATHENA is **not** being built to beat HFT firms on speed.

It is being built to make **better medium-term decisions** using data.

Our edge will come from:

- Combining technical and fundamental signals.
- Consistent risk management.
- Evidence-based strategy testing.
- Automation that removes emotional bias.

- Continuous learning and improvement.

Our time horizon is likely to be:

- Swing trading (days to weeks).
- Position trading (weeks to months).
- Long-term investing (months to years).

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## Mental Model

Imagine a chess tournament.

You don't enter expecting to beat the world champion on your first day.

Instead, you choose games where your preparation gives you an edge.

ATHENA's philosophy is the same:

**Don't compete where others have overwhelming structural advantages. Compete where disciplined research creates an edge.**

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## How This Influences ATHENA's Architecture

Today's lesson directly shapes the platform.

We'll eventually have modules such as:

ATHENA

Market Data



Signal Engine



Strategy Engine



Risk Engine



Portfolio Construction



Backtesting



Performance Analytics

Every module exists to improve decision quality—not trading speed.

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### **40-Minute Practical Exercise**

#### **Part 1 (20 minutes): Observe Market Participants**

Choose **one Indian stock** and **one US stock** that you're familiar with.

For each, answer:

1. Who is most likely buying today?
2. Who is most likely selling today?
3. Is the move driven by news, expectations, or broader market sentiment?
4. Would a mutual fund, hedge fund, or retail investor likely care most about this move?

Write down your reasoning. The goal is not to be "right" but to practice thinking beyond the price chart.

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#### **Part 2 (20 minutes): Start the ATHENA Watchlist**

Create a spreadsheet (or CSV) with these columns:

**Symbol Market Sector Market Cap Why I'm Watching Institutional Interest?**

Add **10 Indian stocks** and **10 US stocks**.

Don't worry about perfect picks. This watchlist will evolve throughout the course and eventually feed into ATHENA's scanner.

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### **Review Questions**

Try answering these without looking back.

1. Why isn't every seller necessarily bearish on a stock?
  2. Why don't we plan to compete with HFT firms?
  3. Name two advantages that an individual investor has over many institutions.
  4. What is ATHENA's intended competitive edge?
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## Key Takeaways

- Every trade has another participant with different motivations.
- Different participants operate under different constraints.
- Retail investors can build genuine advantages through patience and discipline.
- ATHENA is designed to optimize **decision quality**, not execution speed.
- Understanding your competition is essential before designing strategies.